

Read two numbers and perform arithmetic operation using menu with switch:

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
#include<math.h>
```

```
void main()
```

```
{
```

```
float a, b; char op;
```

```
abc:
```

```
clrscr();
```

```
printf("Enter a, b values "); scanf("%f %f",&a, &b);
```

```
puts("-----");
```

```
puts("\t\t\t M E N U");
```

```
puts("-----");
```

```
puts("\t\t\t +. Add");
```

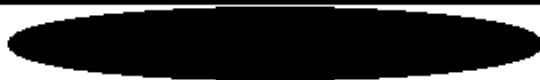
```
puts("\t\t\t -. Sub");
```

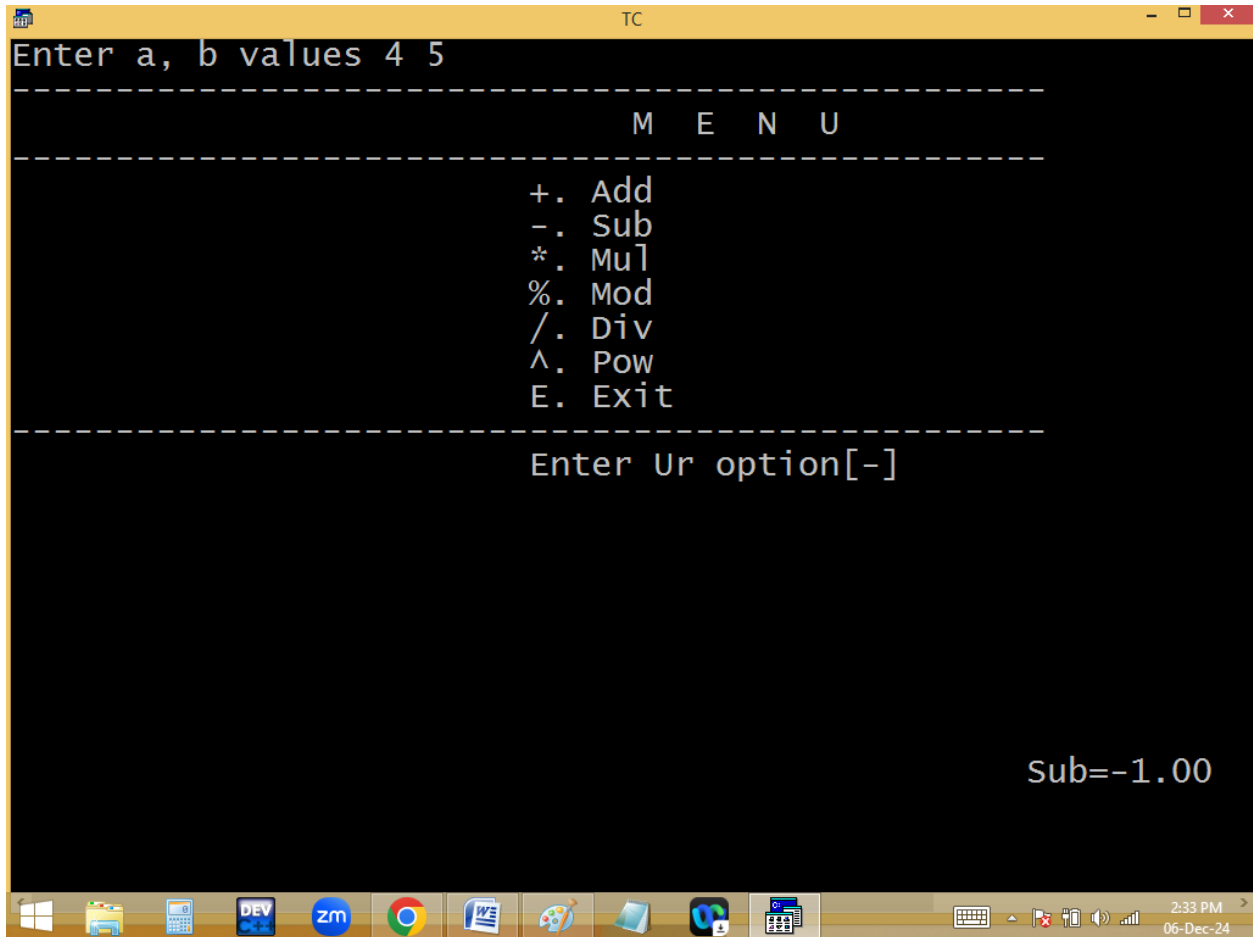
```
puts("\t\t\t *. Mul");
```

```
puts("\t\t\t %. Mod");
puts("\t\t\t /. Div");
puts("\t\t\t ^. Pow");
puts("\t\t\t E. Exit");
puts("-----");
printf("\t\t\t Enter Ur option[ ]\b\b");
flushall();
scanf("%c",&op);
gotoxy(50, 22);
switch(op)
{
case '+': printf("Sum=%.2f",a+b);break;
case '-': printf("Sub=%.2f",a-b);break;
case '*': printf("Mul=%.2f",a*b);break;
case '%': printf("Mod=%.2f",fmod(a,b));break;
case '/': printf("Div=%.2f",a/b);break;
case '^': printf("Pow=%.2f",pow(a,b));break;
```

```
case 'e': case 'E': return;  
default: printf("Invalid Option");  
}  
  
getch();  
goto abc;  
}
```

Enter two numbers 5 9
M E N U
+. Add -. Sub *. Mul %. Mod /. Div ^ . Pow
Enter the operator [*]
Product = 45





```
TC
Enter a, b values 3 4
-----
M E N U
-----
+. Add
-. Sub
*. Mul
%. Mod
/. Div
^. Pow
E. Exit
-----
Enter Ur option[e]
```

LOOPS / ITERATIONS / REPETITIVE STATEMETNS

Loops are used to repeat a block/group of statements continuously until the given condition becomes false.

Loops reduce program size and improves performance.

In loops beginning and ending points are same.

C-Language supports basically 2 types of loops.

- 1. Entry/pre controlled loops.**
- 2. Exit/post controlled loops.**

In entry control loops, condition is tested first and it is true then only statements block is executed.

Under entry control loops we are having

- i. While loop**
- ii. For loop**

In exit control loop, the statements are executed first and later condition is tested.

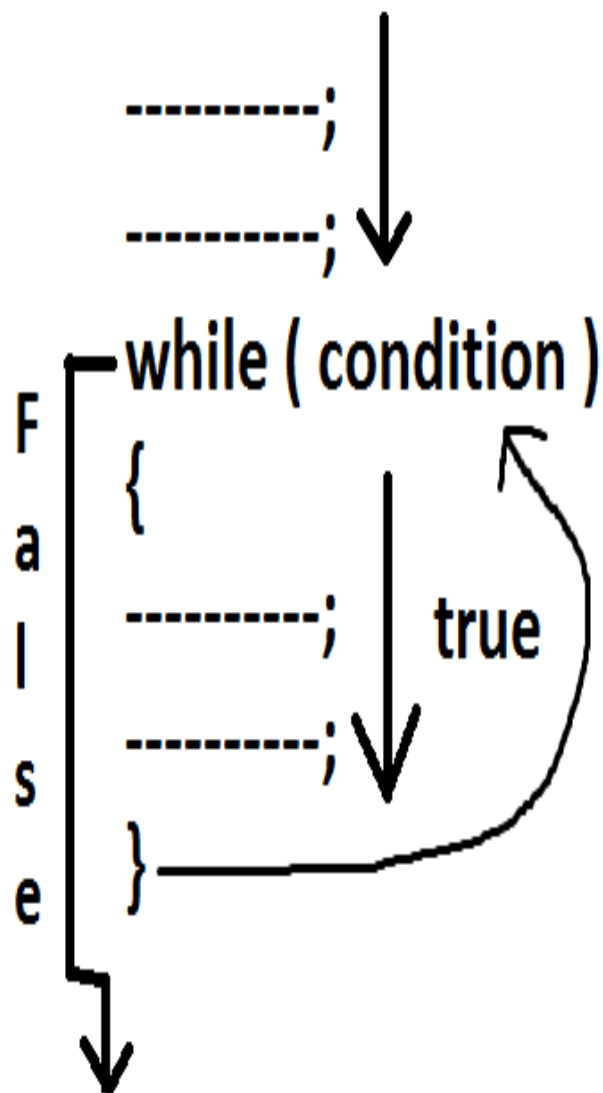
Under exit control loop we are having

i. do while.

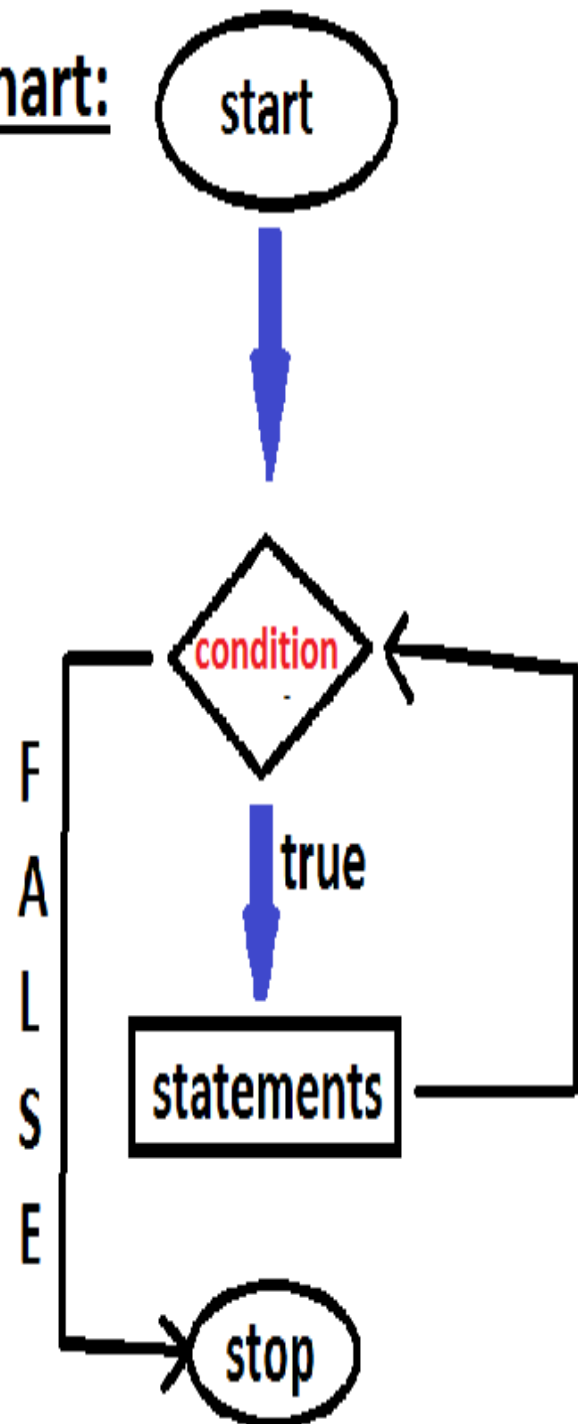
While loop:

- **while is a keyword.**
- **In while loop condition is tested first and it is true then only while block statements are executed. After executing while block statements, the program execution automatically shifted/jumped to while condition at the beginning. If it is true then once again the while block statements are repeated. Like this the process is continued until while condition becomes false.**
- **While is entry control loop.**

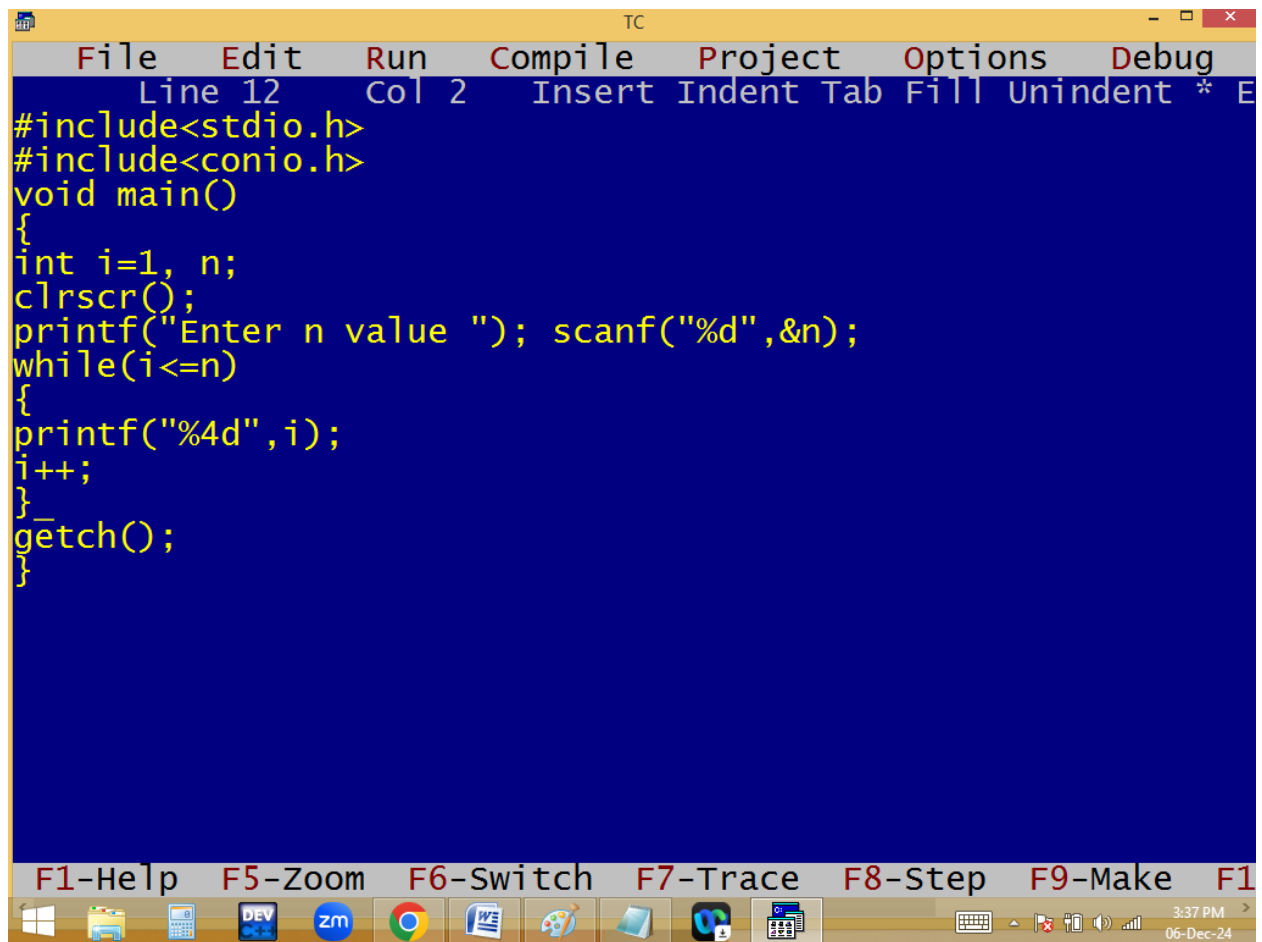
Syntax:



Flow chart:



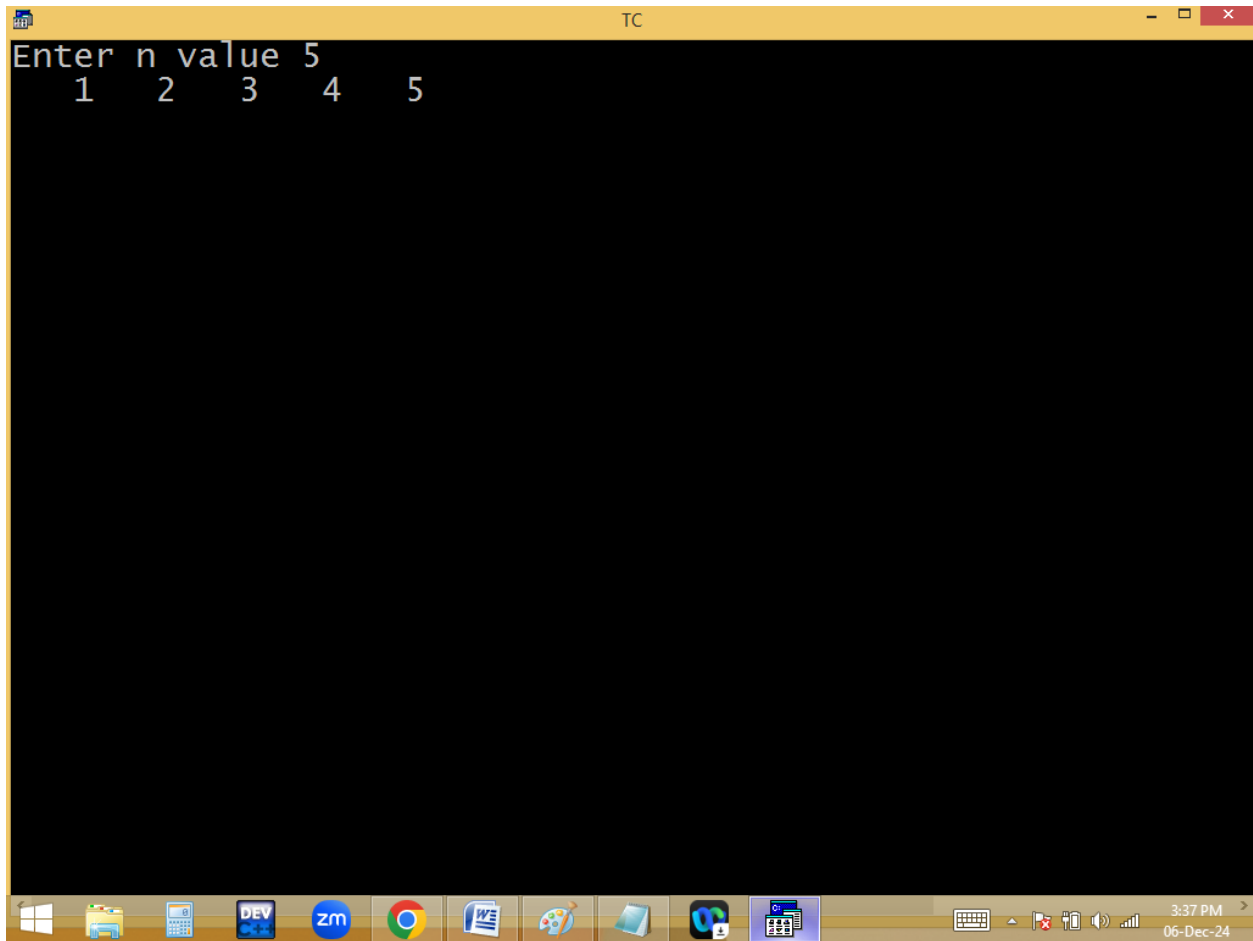
Eg: printing 1..n numbers

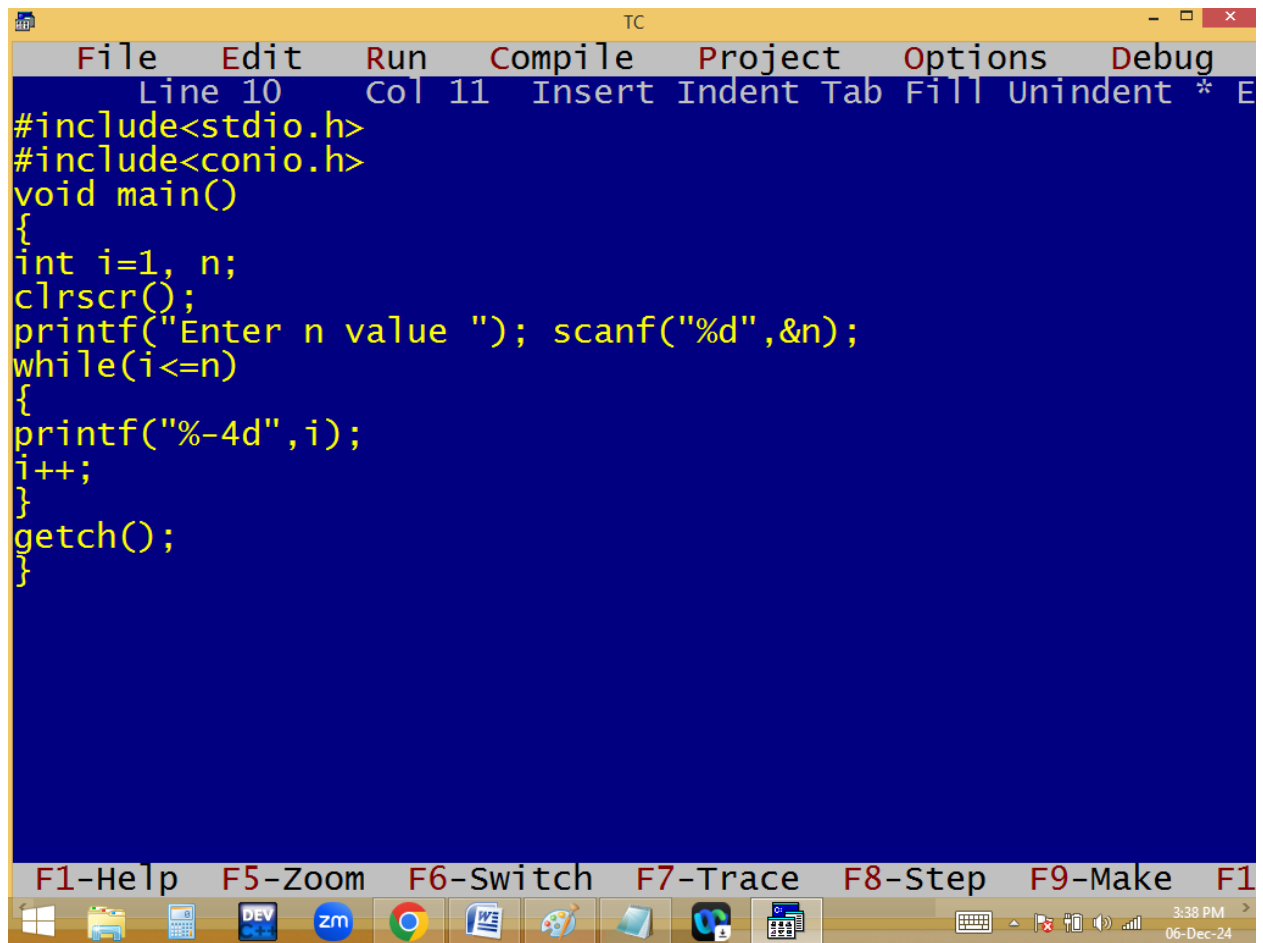


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 12 Col 2" and lists keyboard shortcuts: Insert, Indent, Tab, Fill, Unindent, *, and E. The code editor contains the following C program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int i=1, n;
    clrscr();
    printf("Enter n value "); scanf("%d",&n);
    while(i<=n)
    {
        printf("%4d",i);
        i++;
    }
    getch();
}
```

The bottom of the window features a toolbar with icons for various functions and a status bar at the very bottom showing the time "3:37 PM" and the date "06-Dec-24".





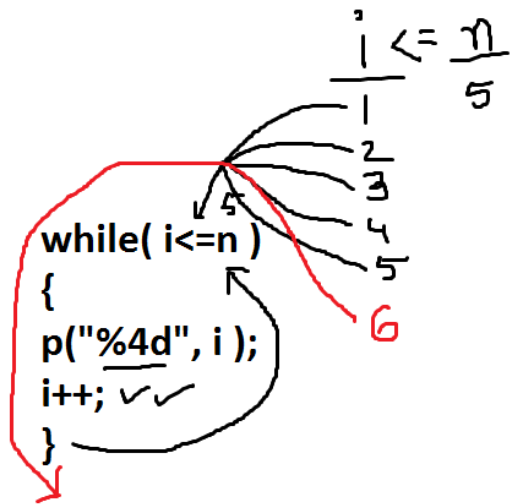
```
TC
File Edit Run Compile Project Options Debug
Line 10 Col 11 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1, n;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(i<=n)
{
printf("%-4d",i);
i++;
}
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit

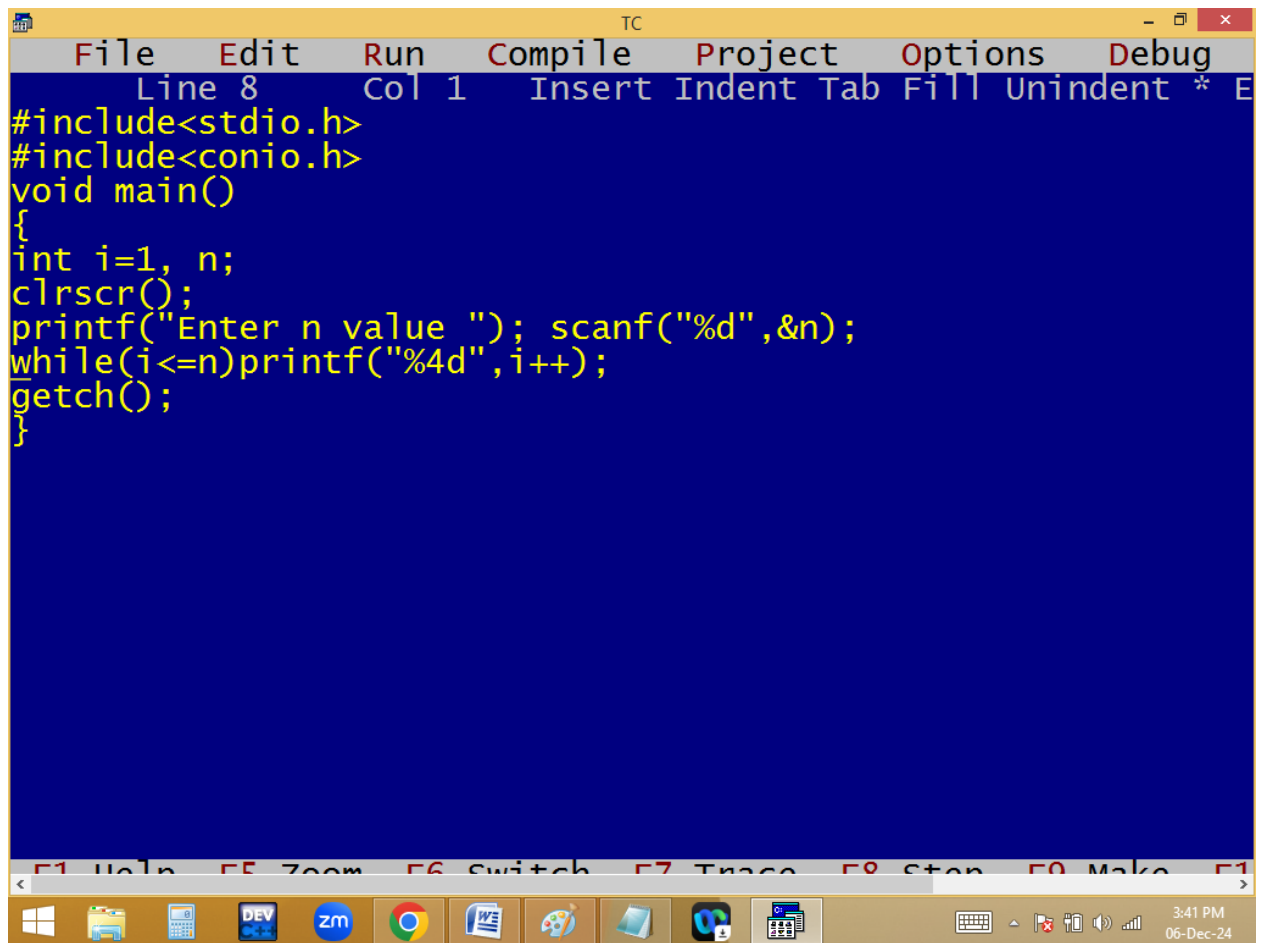
3:38 PM 06-Dec-24

```
TC
Enter n value 5
1 2 3 4 5
```

```
TC
Enter n value 100
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40
41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60
61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80
81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
_
```



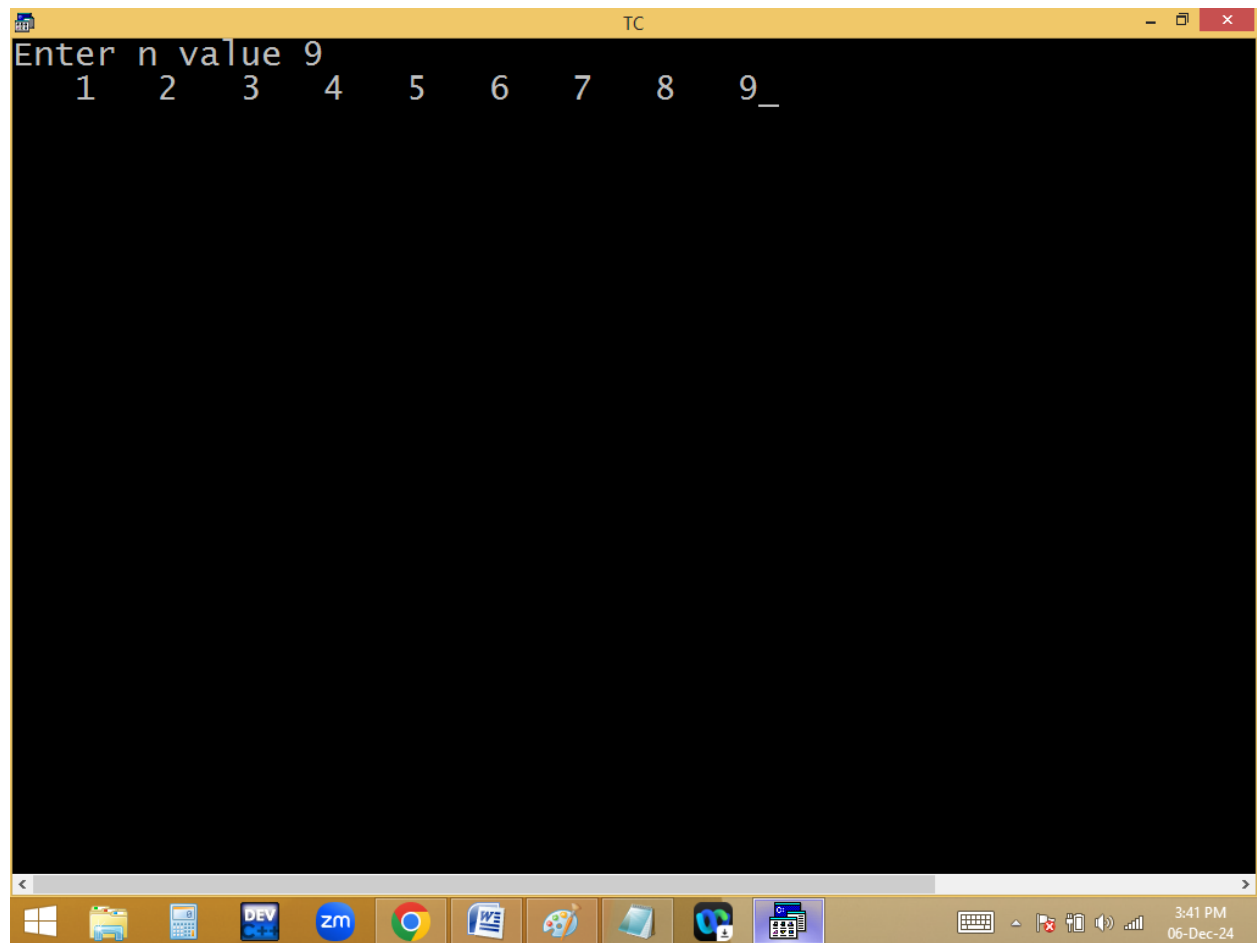
1	2	3
4	5	

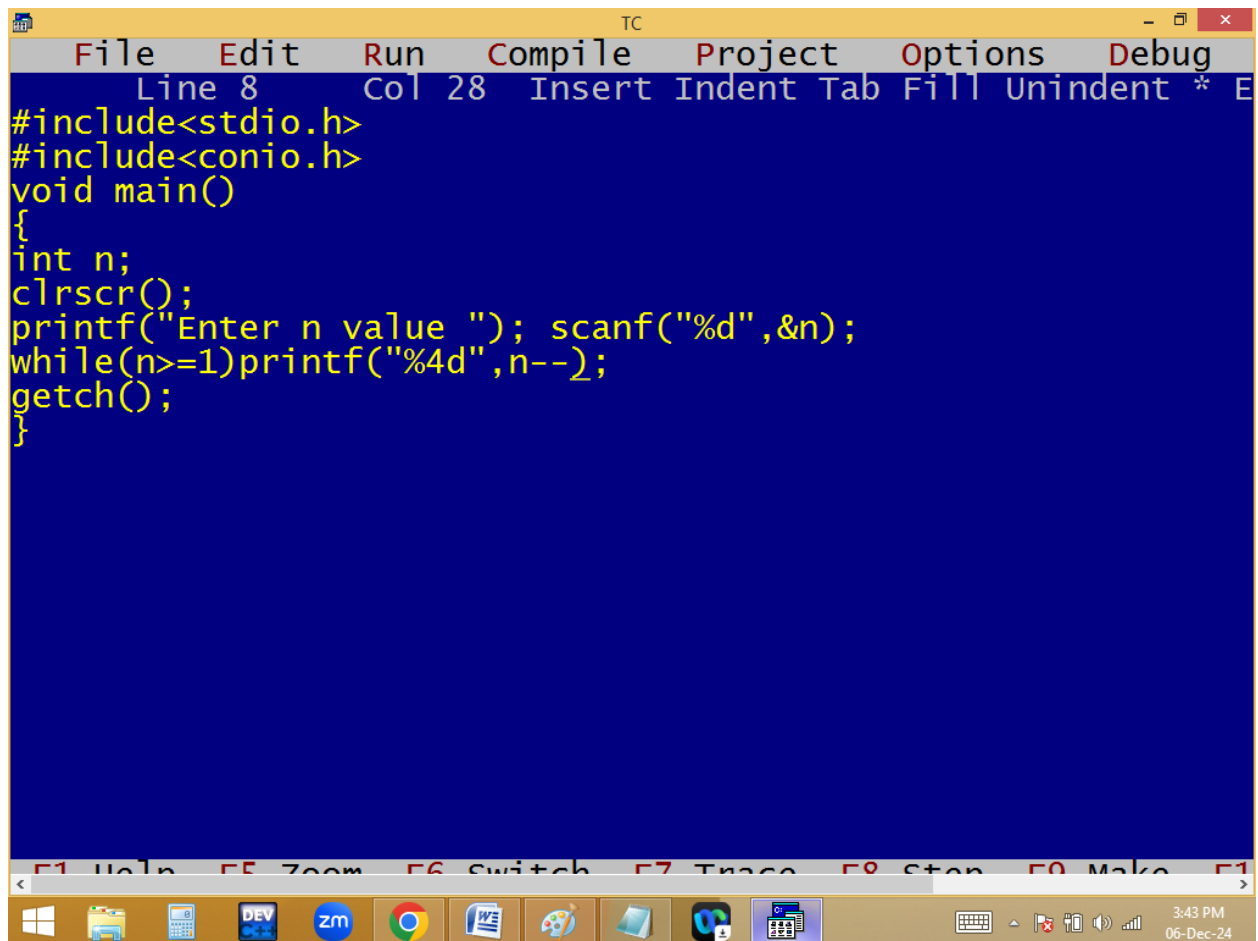


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 8", "Col 1", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1, n;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(i<=n)printf("%4d",i++);
getch();
}
```

At the bottom of the window, there is a toolbar with icons for various functions, and a status bar showing the time "3:41 PM" and the date "06-Dec-24".





```
TC
File Edit Run Compile Project Options Debug
Line 8 Col 28 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(n>=1)printf("%4d",n--);
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

Windows taskbar icons: File Explorer, Calculator, DEV C++, Zoom, Chrome, Word, Paint, File Explorer, OneDrive, Task Manager. System tray: 3:43 PM, 06-Dec-24.

